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Visible-Light-Induced Regioselective Radical-Polar Crossover 1,4-Hydrophosphinylation of 1,3-Enynes: Access to Trisubstituted Allenes Bearing a Phosphine Oxide Group

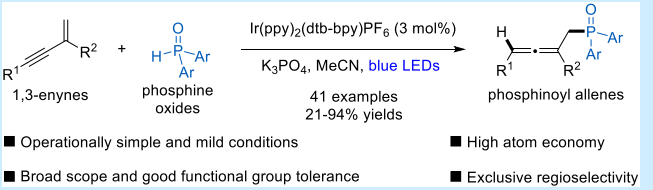
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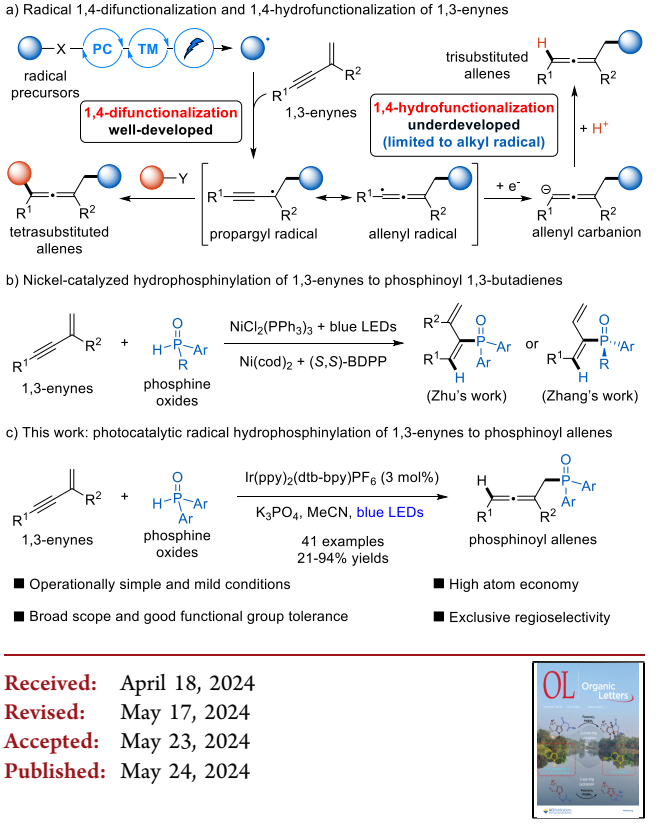
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ABSTRACT: The radical 1,4-functionalizations of 1,3-enynes have emerged as a powerful strategy for the synthesis of multisubstituted allenes. However, the phosphorus-centered radical-initiated transformations remain largely elusive. Herein, visible-light photoredox catalytic regioselective radical hydrophosphinylation of 1,3-enynes with diaryl phosphine oxides as phosphinoyl radical precursors has been realized. This protocol features mild conditions, a wide substrate scope, and good functional group tolerance, producing a diverse range of phosphinoyl-substituted allenes in moderate to good yields with high atom economy. Detailed mechanistic experiments revealed a radical-polar crossover process in the reaction.



Allenes are important structural motifs commonly encountered in bioactive natural products, pharmaceuticals, and functional materials but also represent useful building scaffolds for the synthesis of diverse complex molecules.[1] Consequently, a great deal of effort has been devoted to developing efficient methods for the synthesis of allene compounds.[2] In recent years, the radical 1,4-functionalizations of 1,3-enynes have emerged as a robust platform for the concise and efficient synthesis of multisubstituted allenes (Scheme 1a).[3] Mechanistically, the highly reactive radical intermediates, generated from radical precursors via photocatalysis, transition metal catalysis, or electrochemistry, can efficiently undergo addition to the olefin unit of 1,3-enynes to deliver propargyl radical intermediates, which are capable of undergoing reversible resonance to form allenyl radicals. The sequential regioselective functionalization of allenyl radicals enables the 1,4-difunctionalization of 1,3-enynes while producing tetrasubstituted allenes (Scheme 1a, left).[4] Alternatively, a mechanistically distinct protocol is the radical 1,4-hydrofunctionalization of 1,3-enynes through a radical-polar crossover pathway,[5] in which a single-electron reduction of allenyl radicals forms allenyl carbanion intermediates, followed by the protonation to afford trisubstituted allenes (Scheme 1a, right).[6] Compared with the well-developed 1,4-difunctionalization of 1,3-enynes, the occurrence of radical 1,4-hydrofunctionalization reactions is extremely rare and limited to alkyl radicals.[6] Therefore, the development of a novel and efficient strategy for the radical-mediated hydrofunctionalization of 1,3-enynes, particularly

Scheme 1. Difunctionalization of 1,3-Enynes



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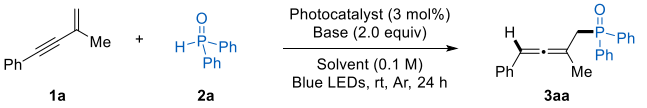
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targeting heteroatom radicals, to access functionalized allenes remains highly desirable.

On the contrary, organophosphorus compounds play significant and versatile roles in catalysis, organic synthesis, medicinal chemistry, and materials sciences.[7] Among them, allenylphosphine derivatives have garnered much synthetic interest because of their extensive applications as synthetic intermediates, ligands, organocatalysts, and biologically active reagents.[8,9] Traditionally, these scaffolds have been prepared by the [2,3]-sigmatropic rearrangement[10] or propargylic substitution reactions,[11] which often suffer from significant limitations, including the necessity for highly prefunctionalized starting materials and unstable and hazardous phosphorus reagents, low atom economy, and poor functional group tolerance. Over the past decade, visible-light-induced photoredox catalysis has emerged as a promising strategy for generating reactive phosphorus radicals under mild conditions through hydrogen atom transfer (HAT) or proton-coupled electron transfer (PCET).[12] In this context, photocatalytic P-centered radical-triggered difunctionalization of unsaturated hydrocarbons has been exploited as a straightforward and efficient strategy for the rapid assembly of a wide range of structurally complex phosphorus-containing compounds.[13] Inspired by these precedents, we hypothesized that photocatalytic hydrophosphinylation reactions of 1,3-enynes with phosphinoyl radicals could offer an ideal synthetic strategy for trisubstituted phosphinoyl-containing alkenes.[14] Very recently, Zhu and co-workers reported a visible-light-driven nickel-catalyzed hydrophosphinylation of the alkyne motif of 1,3-enynes with phosphine oxides to afford various phosphinoyl 1,3-butadienes.[15a] Additionally, Zhang's group achieved asymmetric induction in this reaction by employing a chiral diphosphine ligand (Scheme 1b).[15b] These findings may indicate that achieving regioselectivity in our proposed photocatalytic radical hydrophosphinylation strategy poses significant challenges. Herein, we present a regioselective hydrophosphinylation of 1,3-enynes with diarylphosphine oxides under visible-light-induced photoredox catalysis (Scheme 1c). This reaction provides a straightforward synthetic approach to diverse phosphinoyl-substituted allenes in moderate to good yields with high atom economy and a broad substrate scope.

To test the feasibility of our hypothesis, we chose (3-methylbut-3-en-1-yn-1-yl)benzene (**1a**) and diphenylphosphine oxide (**2a**) as model substrates (Table 1). In the presence of $Ru(bpy)_3Cl_2·6H_2O$ (3 mol %) as a photocatalyst and K_3PO_4 (2.0 equiv) as a base, the hydrophosphinylation proceeded smoothly in MeCN under 40 W blue light-emitting diode (LED) irradiation, affording the desired phosphinoyl allene product **3aa** in 43% isolated yield with excellent regioselectivity (entry 1). Then, we employed alternative metal-based photocatalysts and organic dyes for this reaction (entries 2−4), and $Ir(ppy)_2(dtb\text{-}bpy)PF_6$ delivered the best product yield (81%). Next, various inorganic and organic bases such as K_2CO_3, Cs_2CO_3, and DBU were screened (entries 5−7, respectively), but no improvement of the results was observed. The reaction was remarkably sensitive to the choice of reaction media, with no product obtained when alternative solvents such as DMF, DCM, and MeOH were employed (entries 8−10, respectively). Notably, reducing the loading of 1,3-enyne substrate **1a** (1.3 equiv) increased the yield to 85%, while a lower reaction efficiency was observed with 1 mol % photocatalyst (entries 11 and 12, respectively). Finally, control

Table 1. Optimization of the Reaction Conditions[a]



entry	photocatalyst	base	solvent	yield (%)[b]
1	$Ru(bpy)_3Cl_2·6H_2O$	K_3PO_4	MeCN	43
2	$Ir(ppy)_2(dtb\text{-}bpy)PF_6$	K_3PO_4	MeCN	81
3	*fac*-$Ir(ppy)_3$	K_3PO_4	MeCN	32
4	4CzIPN	K_3PO_4	MeCN	40
5	$Ir(ppy)_2(dtb\text{-}bpy)PF_6$	K_2CO_3	MeCN	37
6	$Ir(ppy)_2(dtb\text{-}bpy)PF_6$	Cs_2CO_3	MeCN	51
7	$Ir(ppy)_2(dtb\text{-}bpy)PF_6$	DBU	MeCN	44
8	$Ir(ppy)_2(dtb\text{-}bpy)PF_6$	K_3PO_4	DMF	trace
9	$Ir(ppy)_2(dtb\text{-}bpy)PF_6$	K_3PO_4	DCM	0
10	$Ir(ppy)_2(dtb\text{-}bpy)PF_6$	K_3PO_4	MeOH	0
11[c]	$Ir(ppy)_2(dtb\text{-}bpy)PF_6$	K_3PO_4	MeCN	85
12[c,d]	$Ir(ppy)_2(dtb\text{-}bpy)PF_6$	K_3PO_4	MeCN	75
13[c]	–	K_3PO_4	MeCN	0
14[c]	$Ir(ppy)_2(dtb\text{-}bpy)PF_6$	–	MeCN	trace
15[c,e]	$Ir(ppy)_2(dtb\text{-}bpy)PF_6$	K_3PO_4	MeCN	0

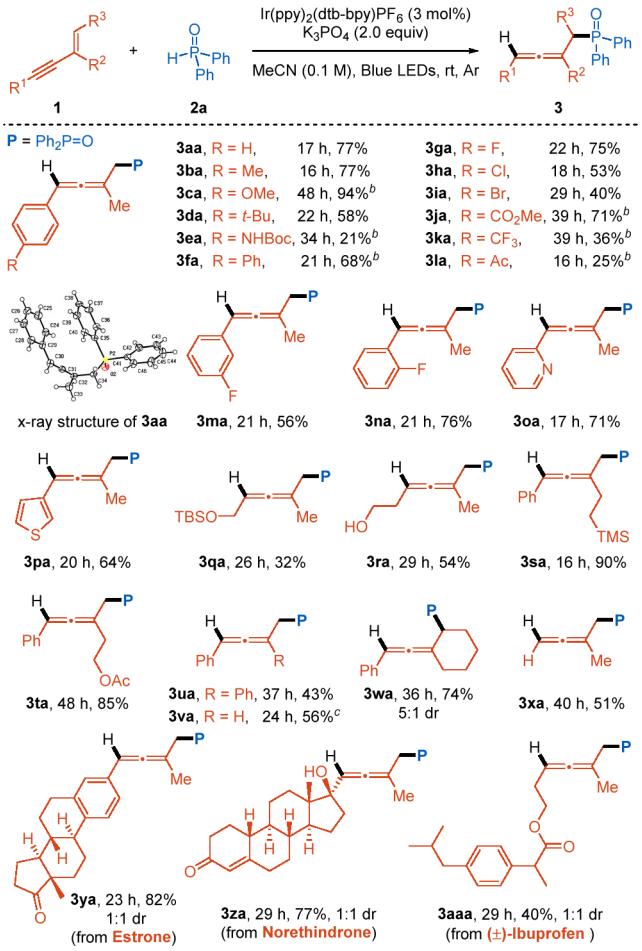
[a]Unless otherwise noted, reactions were carried out with **1a** (0.3 mmol, 1.5 equiv), **2a** (0.2 mmol), a photocatalyst (3 mol %), a base (2.0 equiv), and a solvent (2 mL) and irradiation via a 40 W blue LED under Ar at room temperature for 24 h. [b]Isolated yield. [c]**1a** (0.26 mmol, 1.3 equiv) was used. [d]$Ir(ppy)_2(dtb\text{-}bpy)PF_6$ (1 mol %) was used. [e]The reaction was performed in the dark.

experiments demonstrated that the photocatalyst, base, and visible-light irradiation were essential for this reaction (entries 13−15, respectively).

With the optimized conditions in hand, we then explored the scope of 1,3-enynes with diphenylphosphine oxide **2a** as the reaction partner (Scheme 2). A wide variety of 2-methyl-4-aryl-substituted 1,3-enynes with an electron-donating or -withdrawing group at different positions of the benzene ring smoothly participated in this reaction to give the desired trisubstituted allenes **3aa**−**3na** in moderate to good yields. The precise structure of product **3aa** was further confirmed by X-ray crystallographic analysis (CCDC 2311675). The substrates bearing heterocycles such as pyridine and thiophene were found to be compatible with the present reaction conditions to produce products **3oa** and **3pa** in 71% and 64% yields, respectively. Notably, 2,4-dialkyl-substituted 1,3-enynes also proved to be suitable substrates for this hydrophosphinylation reaction. In addition, the methyl group at position 2 of 1,3-enynes could be replaced with other functionalized alkyl and phenyl groups and hydrogen, producing the desired allenes **3sa**−**3va** in good yields. Moreover, a series of functional groups, such as ether (**3ca**), amide (**3ea**), esters (**3ja** and **3ta**), ketone (**3la**), hydroxyl (**3ra**), and silyl (**3qa** and **3sa**) groups, were very well tolerated. In addition to terminal 1,3-enynes, the cyclic internal substrate was amenable to this transformation, delivering the desired product **3wa** in 74% yield with a high diastereoselectivity (5:1 dr). We also found that 2-methylbut-1-en-3-yne **1x** was an effective substrate for producing 1,1-disubstituted allene product **3xa** in 51% yield. To demonstrate the potential applicability of this protocol, we examined more complex substrates for the hydrophosphinylation reaction. 1,3-Enynes derived from estrone, norethindrone, and (±)-ibuprofen were well tolerated and afforded the corresponding allene derivatives smoothly.

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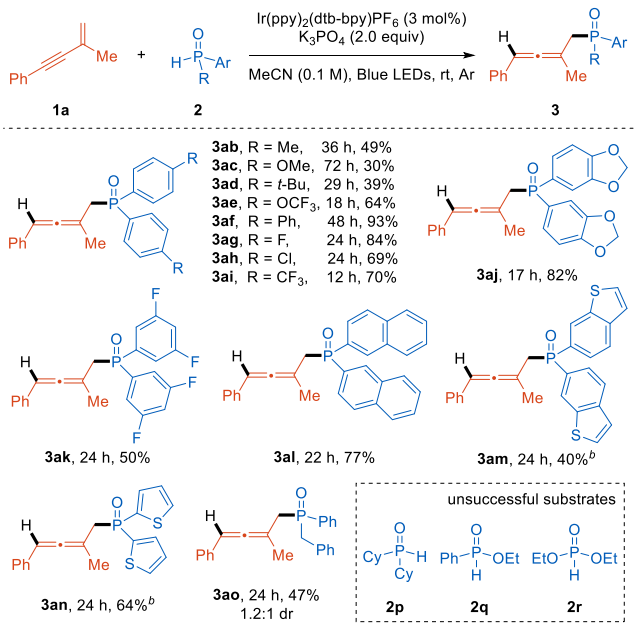
Scheme 2. Scope of 1,3-Enynes[a]



[a]Reaction conditions: **1** (0.52 mmol, 1.3 equiv), **2a** (0.4 mmol), $Ir(ppy)_2(dtb\text{-}bpy)PF_6$ (0.012 mmol, 3 mol %), K_3PO_4 (0.8 mmol, 2.0 equiv), MeCN (4.0 mL), 40 W blue LEDs, Ar, rt. [b]Cs_2CO_3 (0.2 equiv) was used as the base. [c]DABCO (2.0 equiv) was used as the base.

Next, the scope of secondary phosphine oxide was investigated. As shown in Scheme 3, a variety of diarylphosphine oxides smoothly participated in this reaction. Incorporation of either an electron-donating or -accepting substituent on the benzene ring of diarylphosphine oxides was compatible with the reaction conditions, providing phosphinoyl-substituted allene products **3ab**−**3ak** in moderate to good yields. It is worth noting that electron-deficient diarylphosphine oxides (**3ag**−**3ai**) generally gave a reaction efficiency higher than those seen with electron-rich ones (**3ab**−**3ae**), perhaps because electron-rich secondary phosphine oxides exist nearly exclusively in the pentavalent tautomeric form to prevent the photocatalytic generation of phosphinoyl radicals.[13a,16] Moreover, polycycle- and heterocycle-substituted diarylphosphine oxides were suitable substrates for producing the corresponding products **3al**−**3an** in 40−77% yields. Note that the reaction of benzylphenylphosphine oxide **2o** also proceeded smoothly to afford the desired product **3ao** in moderate yield, while dicyclohexylphosphine oxide **2p**, ethyl phenylphosphinate **2q**, and diethyl phosphite **2r** failed to participate in the hydrophosphinylation reaction presented here at the current stage.

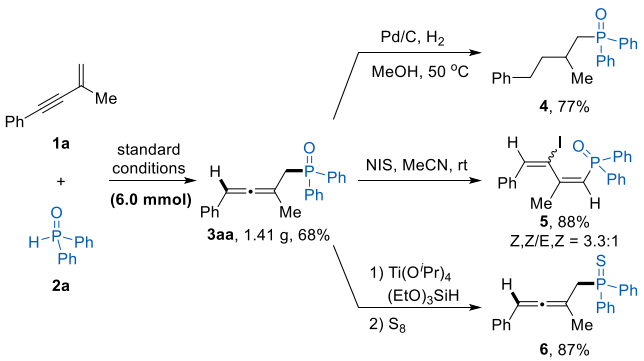
Scheme 3. Scope of Secondaryl Phosphine Oxides[a]



[a]Reaction conditions: **1a** (0.52 mmol, 1.3 equiv), **2** (0.4 mmol), $Ir(ppy)_2(dtb\text{-}bpy)PF_6$ (0.012 mmol, 3 mol %), K_3PO_4 (0.8 mmol, 2.0 equiv), MeCN (4.0 mL), 40 W blue LEDs, Ar, rt. [b]DABCO (2.0 equiv) was used as the base.

To further demonstrate the synthetic utility of this protocol, a gram-scale model reaction (6 mmol) was carried out under the optimal conditions, and 1.41 g of the desired product **3aa** was obtained in 68% yield (Scheme 4). In addition, the

Scheme 4. Utility of the Reaction Methodology



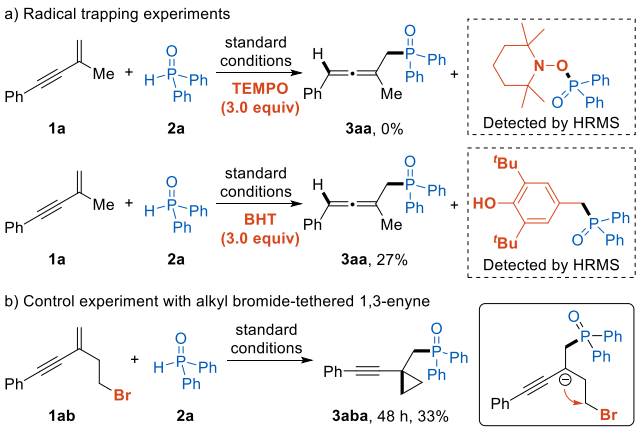
phosphinoyl-substituted allene products obtained by this reaction can serve as valuable building blocks for further transformations. The simultaneous reduction of two double bonds of allene **3aa** with a catalytic amount of Pd/C under a hydrogen gas atmosphere afforded diphenyl alkyl phosphine oxide **4** in 77% yield. Treatment of **3aa** with *N*-iodosuccinimide (NIS) led to iodo-1,3-diene **5** in good yield through allenylic C−H bond elimination. Furthermore, the phosphine oxide group in **3aa** could be readily reduced by Si−H reagent and then oxidized in the presence of elemental sulfur to afford phosphine sulfide **6** in 87% yield.

To gain some insight into the mechanism of this photocatalytic hydrophosphinylation reaction, a series of control experiments were conducted. Initially, the radical-trapping experiments with 2,2,6,6-tetramethyl-1-piperidinooxy

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(TEMPO) and 2,6-di-*tert*-butyl-4-methylphenol (BHT) showed an obvious suppression of the desired transformation, and the adducts of the phosphinoyl radical by TEMPO and BHT were confirmed by high-resolution mass spectrometry (HRMS), which indicated that the reaction might proceed through a phosphorus radical pathway (Scheme 5a). Next, we

Scheme 5. Mechanistic Investigations

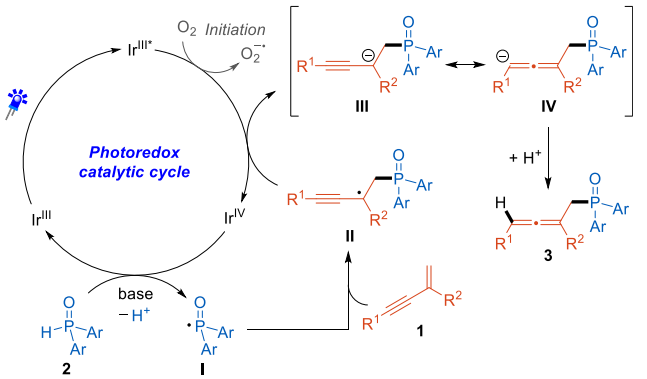


also performed several deuterium labeling experiments to identify the hydrogen donor for this reaction (see Figures S6–S8). The reaction of 1,3-enyne **1a** and deuterated **2a**-*d* (80% D) produced deuterated product **3aa**-*d* with 29% deuterium content at the allenyl position. Replacing MeCN with d_3-MeCN resulted in deuterated product **3aa**-*d* in 61% yield with 23% D. When 5 equiv of D_2O was added to the model reaction mixture, deuterated product **3aa**-*d* was afforded in 79% yield with 70% D, and the ratio of deuterium in **3aa**-*d* increased accordingly with the D_2O loading. These results indicated that the hydrogen atom at the allenyl position in the product mainly originated from protons in the reaction system. In addition, the control experiment with alkyl bromide-tethered 1,3-enyne **1ab** as the substrate gave alkynylcyclopropane product **3aba** via intramolecular nucleophile substitution (Scheme 5b),[17] which provided strong support for an allenyl carbanion being involved as the key intermediate in this reaction. Finally, light on/off experiments demonstrated that continuous light irradiation was essential for the progress of this protocol (Figure S9).

On the basis of the mechanistic studies presented above and related literature reports,[3−6,12−14] a radical-polar crossover mechanism was proposed (Scheme 6). First, we postulated that excited photocatalyst Ir^{III}* would undergo a single-electron oxidation by the residual oxygen in this reaction system,[18] to initiate the first catalytic cycle and provide oxidized photocatalyst Ir^{IV}. This strong oxidative photocatalyst ($E_{1/2}^{IV/III}$ = +1.2 V vs SCE)[18] can oxidize diarylphosphine oxide **2** (E^{ox} = +1.0 V vs SCE for **2a**)[13e] to generate phosphinoyl radical intermediate **I** via PCET with the assistance of a base. Then, the regioselective addition of radical **I** to 1,3-enyne **1** affords propargyl radical intermediate **II**. The subsequent SET from excited photocatalyst Ir^{III}* to propargyl radical **II** produces propargyl carbanion **III**, which can undergo reversible resonance to give trisubstituted allenyl carbanion **IV**. Finally, protonation of the allenyl carbanion delivers the desired phosphinoyl-substituted allene product **3**.

In summary, we have developed an efficient radical hydrophosphinylation of 1,3-enynes with secondary phosphine

Scheme 6. Proposed Reaction Mechanism



oxides via visible-light photoredox catalysis. Under simple operation and mild conditions, various trisubstituted allenes bearing a phosphine oxide group were afforded in moderate to good yields with high atom economy and excellent regioselectivity. This method can be readily scalable, and the phosphinoyl-substituted allene products could serve as valuable building blocks for further transformations. The results of the deuteration and control experiments suggested that a radical-polar crossover process should account for allene generation.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.4c01422.

Experimental procedures and spectral data of all new compounds (PDF)

Accession Codes

CCDC 2311675 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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